

Closed-Form Solution for Defaultable Bond Options under a Two-Factor Gaussian Model for Risky Rates Modeling

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Takeaways

- An option pricing model that considers the dynamic of risky rates under a two-factor Gaussian model where risky rates are the sum of risk-free rates and credit spreads is proposed. This distributional hypothesis is consistent with negative interest rates and benefits from analytical tractability.
- Under this framework, closed formulas for the price of European options having as their underlying both defaultable zero-coupon bonds and coupon-bearing bonds are derived under an approximately log-normal distribution assumption.
- A detailed two-step calibration procedure is provided. The first step estimates the parameters of the one-factor Hull-White model for the risk-free rates and the second step estimates the parameters of the model's credit component under a grid of correlation values.

Abstract

In this paper, we provide a closed-form solution for defaultable bond options under a two-factor Gaussian model for risky rates. The key feature of the proposed stochastic model is the introduction of two stochastic dynamics to address the behavior of both risk-free interest rates and credit spreads where the two sources of risk are correlated. Moreover, the model is able to match exactly the term structure of defaultable interest rates. In order to model credit-sensitive bonds, we assume an intensity-based reduced-form framework where the approach based on the recovery of market value is considered and where the recovery rate is one of the model's input. Under this framework, we derive closed formulas for the price of European options having as their underlying both defaultable zero-coupon bonds and coupon-bearing bonds which we assume are approximately log-normally distributed. Moreover, the options are assumed to be knocked out upon a default event of the bond with zero value to the option holder. An empirical analysis is performed to illustrate the calibration process of the proposed model using financial market data to price calls and puts options across different credit ratings and different levels of the loss given default.

The computation of quantitative metrics for fixed income securities in order to price and quantify their risks requires the modeling of both risk-free and risky interest rates. There is an extensive literature dealing with the modeling of risk-free interest rates. In particular, for interest rate derivatives such as caps/floors, swaptions, and bond options, closed-form solutions are available for pricing purposes. Such formulas are widely used by practitioners to evaluate, quote, and trade, in a very efficient way, fixed income instruments in the over-the-counter market. However, in the case of options written on bonds, it is critical to take into account that bond pricing is affected by risky interest rates, namely, by both risk-free interest rates and credit spreads. Unfortunately, because the market for options on defaultable bonds is not particularly liquid, having at disposal modeling techniques for fixed-income instruments is essential for pricing and hedging purposes.

In pricing credit risky fixed-income instruments, a pragmatic approach might be to ignore the credit risk and price the options as default-free interest rate options. This, however, is inconsistent with the absence of arbitrage and the existence of spreads between risky and risk-free interest rates. An alternative approach is to apply the structural model framework formulated by Merton (1974) who assumes that a firm's value follows a stochastic process and that default occurs when the firm's value hits a predefined lower barrier. This approach to valuation has been and widely used in credit risk modeling. However, because the firm's capital structure is generally complex, using this approach to price credit-sensitive instruments is difficult to implement.

The most effective approach for pricing and hedging of credit-sensitive instruments seems to be the intensity-based reduced-form framework. This approach assumes that the default of a firm is an exogenous stochastic event that is independent of the firm's assets and its capital structure. There are two approaches suggested in the literature for the intensity-based reduced-form framework for valuing credit-sensitive instruments. The first approach — suggested by Lando (1994), Das and Tufano (1996), Jarrow et al. (1997), and Lando (1998) — models default as a transition to an absorbing state of a Markov chain. The second approach — proposed by Jarrow and Turnbull (1995), Madan and Unal (1998), Lando (1998), Duffie and Singleton (1999) and Collin-Dufresne et al. (2004) — models the times of default as jump times of a Poisson process.

The intensity-based reduced-form framework provides a general framework for pricing defaultable credit-instrument pricing without specification of the functional form for both interest rate and default intensity processes. Hence, they do not provide solutions that are directly implementable. To fill this gap in the pricing issue, some researchers have focused on intensity-based reduced-form approach proposed closed-form solutions for credit-sensitive instruments. Schombucher (2004) has enriched the intensity-based reduced-form framework used for the pricing of credit derivatives by dealing with realistic dynamics of defaultable bond prices.

Tchuindjo (2007) assumes the risk-free interest rate and the credit spread are driven by a two correlated model developed by Hull and White (1990) and proposes closed-form solutions for valuing credit-risky bonds under the recovery of market value assumption of Duffie and Singleton (1999). Tchuindjo (2011) proposes closed-form solutions for pricing European options written on defaultable zero-coupon bonds assuming that both the risk-free interest rate and the credit spread are driven by two correlated models of Ho and Lee (1986), and assuming a zero recovery rate.

The fixed income market has by far fewer zero-coupon bonds with credit risk than coupon-bearing bonds with credit risk. Moreover, the pricing of zero-coupon bonds is impacted by the original discount rules for taxes purpose. So although the existing literature has provided closed-form solutions for options on defaultable zero-coupon bonds, there are no studies that have investigated closed-form solutions for pricing options on defaultable coupon-bearing bonds which is of much more importance to participants in the fixed-income market. Furthermore, previous studies do not provide closed-form solutions where the correlation between risk-free rates and credit spreads are different from zero and where, at the same time, the recovery rate is different from zero. In these cases, typically Monte Carlo simulation, tree implementation or numerical partial differential equations have to be adopted to solve the pricing problem as in Das and Sundaram (2000), Schonbucher (2002), Hua et al. (2009), and Jansen et al. (2018).

In this paper, we provide a closed-form solution for defaultable bond options under a two-factor Gaussian model for risky rates. The key feature of the proposed stochastic model is the introduction of two stochastic dynamics to address the behavior of both risk-free interest rates and credit spreads where the two sources of risk are correlated. Moreover, the model is able to match exactly the term structure of defaultable interest rates. In order to model credit-sensitive bonds, we assume an intensity-based reduced-form framework where the approach based on the recovery of market value is considered and where the recovery rate is one of the model's input. Under this framework, we derive closed formulas for the price of European options having as their underlying both defaultable zero-coupon bonds and coupon-bearing bonds which we assume are approximately log-normally distributed. Moreover, the options are assumed to be knocked out upon a default event of the bond with zero value to the option holder. An empirical analysis has been performed to illustrate the calibration process of the proposed model using financial market data to price calls and puts options across different credit ratings and different levels of the loss given default.

In the next section, we present the theoretical framework for modeling risky rates and then describe the two-factor Gaussian model for risky short-rates. We then focus on pricing and stochastic dynamic for defaultable bonds under the two-factor Gaussian model providing closed-form solutions for options on defaultable bonds. The model's calibration approach under a market-consistent approach is

then described followed by the the results of the empirical analysis.

The framework for modeling risky rates

In the context of risk-free interest rate modeling, short-rate models are widely used. Under this type of model, the instantaneous short-rate, which we will denote by $r(t)$, is modeled on a one-dimensional basis and allows one derive the entire yield curve and its evolution.

Under stochastic interest rates, the price $P(t, T)$ at time t of a risk-free zero-coupon bond with maturity at time $T > t$ is

$$P(t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T r(u) du} \middle| \mathcal{R}_t \right], \quad (1)$$

where $\mathbb{E}^{\mathbb{Q}}$ is the expectation under the risk-neutral measure $\mathbb{Q}$ and $\mathcal{R}_t$ is the sigma-field generated by the term structure of the risk-free rates up to the time t .

In the context of the intensity-based reduced-form framework, the same techniques developed for risk-free rates have been applied to risky rates. In particular, reduced-form models based on an exogenous specification of the dynamics of the default probabilities of the relevant issuer are used under a stochastic intensity framework.

In this paper, we adopt such techniques referring essentially to the framework established in Lando (1998) in relation to the Cox processes (also known as doubly stochastic Poisson processes). Moreover, we apply the recovery of market value (RMV) assumption developed by Duffie and Singleton (1999).

Under this framework, the recovery payment is assumed to be a fraction of the market value of the bond immediately before default and is paid at default. In other words, default implies that a fraction of the market value of the bond is lost. This framework allows us to (1) model the correlation between risk-free interest rates and credit spreads, (2) treat in a realistic way the recovery rate, and (3) benefits from a certain analytical tractability in developing credit models.

Adopting the framework described above under the risk-neutral measure $\mathbb{Q}$, the price $\bar{P}(t, T)$ at time t of a defaultable zero-coupon bond with maturity at time T is

$$\bar{P}(t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T \bar{r}(u) du} \middle| \bar{\mathcal{R}}_t \right], \quad (2)$$

where $\bar{r}(t)$ is the risky or defaultable short-rate and $\bar{\mathcal{R}}_t$ is the sigma-field generated by the term structure of the risky rates up to the time t . The risky short-rate can be computed as,

$$\bar{r}(t) = r(t) + s(t), \quad (3)$$

where $s(t)$ is the short-term credit spread related to a particular issuer. It represents the expected loss rate because it is a measure of the expected loss due to default. Assuming a constant recovery rate denoted by RR , and consequently, a constant loss given default denoted by L , with $RR = 1 - L$, we have that,

$$s(t) = L\lambda(t), \quad (4)$$

where $\lambda(t)$ is the default intensity or hazard rate.

The default intensity at time t captures the conditional default probability in a short-term period following time t . Under the risk-neutral measure $\mathbb{Q}$ and the sigma-field $\mathcal{S}_t$, with a stochastic cumulative default intensity, the survival probability in an interval $[t, T]$, conditional on no default up to time t , is

$$S(t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T \lambda(u) du} \middle| \mathcal{S}_t \right]. \quad (5)$$

with $D(t, T) = 1 - S(t, T)$ being the probability of default in the same interval.

Consequently, following Duffie and Singleton (1999), the price $\bar{P}(t, T)$ at time t of a defaultable zero-coupon bond with maturity at time T can be expressed as follows,

$$\bar{P}(t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T [r(u) + s(u)] du} \middle| \bar{\mathcal{H}}_t \right] = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^T [r(u) + L\lambda(u)] du} \middle| \bar{\mathcal{H}}_t \right]. \quad (6)$$

A two-factor Gaussian model for the risky short-rates

Under the intensity-based reduced-form framework with the assumption of a recovery of market value, we assume a time-inhomogeneous affine model of the term structure of interest rates. These types of models are able to fit the observable term structure, replacing model parameters with a time-dependent function so that the resulting model is time-inhomogeneous.

In particular, we assume a two-factor Gaussian model for risky short-rates that is able to fit exactly the term structure of the risky rates and where the risk-free rates and the credit-spreads are driven by two correlated Gaussian processes.¹

¹There are two possible approaches in modelling risky yields: 1) considering directly the risky yield or 2) modelling the risk-free rate and the credit spread separately. When pricing is concerned, the risky yield is modelled disentangling the risk-free rate and the credit spread dynamics. Treating in a separate way the risk-free rate and the credit spread seems the most frequently used approach for pricing purposes in a market-consistent setting. This is extremely important as it allows one to consider the correlation parameter between the risk-free rate and the credit spread, and the selection of the appropriate recovery rate to be used in the pricing formula. Related examples in the literature are Brigo and Mercurio (2006) where the risk-free rate and the credit spread are treated in a separate way, and Bertocchi et al. (2005) where the risky yield is directly modelled.

Under the proposed model, we assume that LIBOR/swap rates represent the risk-free rates.

Assuming a two correlated Gaussian processes implies that both the risk-free rate and credit-spread can be negative. Dealing with negative risk-free rates is a realistic assumption given that, since 2007, central banks of the major developed countries have pursued a monetary policy to keep short-term borrowing rates very low or negative. Currently, also negative spreads are observed in the market where a possible explanation is related to asset's specific liquidity factors.² However, negative spreads do not imply that probabilities of default are negative, but that the relationship between probability and spread must be enriched by additional factors (i.e., liquidity risk).

In this paper, we assume that spreads embed such additional factors and they can become overall negative. Consequently, intensity is not a pure default intensity but reflects other risk that is different from credit risk, including liquidity risk.³ The reasons reported above support our choice for assuming a Gaussian model, considering also the advantage of its analytical tractability.

The model we propose is such that,

$$\bar{r}(t) = r(t) + s(t), \quad (7)$$

$$r(t) = \alpha(t) + x(t), \quad (8)$$

$$s(t) = L\lambda(t), \quad (9)$$

$$\lambda(t) = \beta(t) + y(t). \quad (10)$$

where $\alpha(t)$ and $\beta(t)$ are deterministic functions of time that permit the fitting of the term structure of risk-free interest rates and the term structure of the credit spreads, respectively, while L is the loss given default which is assumed to be constant. The state variables $x(t)$ represent the stochastic component of the process related to the risk-free short-rates while $y(t)$ is the stochastic component of the process related to the default intensity $\lambda(t)$ and, consequently, of the short-rate credit spread.

As risky interest rates are directly observable in the market, and in order to simplify the model's structure, the two-factor Gaussian model we propose can be expressed in a slightly different form,

$$\bar{r}(t) = \bar{\alpha}(t) + x(t) + Ly(t), \quad (11)$$

where the quantity $\bar{\alpha}(t) = \alpha(t) + L\beta(t)$ is the deterministic function of time that allows us to fit the overall term structure of risky interest rates. This formulation of the model allows us to deal with the variable $\bar{\alpha}(t)$ in lieu of managing both $\alpha(t)$

²See Bhanot and Guo (2011) for more details on this topic.

³It is important to highlight that, in order to calculate the pure default or survival probability under the proposed model, market spreads should be disentangled to identify and quantify their determinants. However, this is beyond the scope of this paper.

and $\beta(t)$. So, in the remainder of this paper, we will refer only to $\bar{\alpha}(t)$ to specify additional model features.

Under this model representation, the state variable $x(t)$ is such that

$$dx(t) = -a_x x(t)dt + \sigma_x dW_x(t), \quad x(0) = 0, \quad (12)$$

where a_x and σ_x are the model parameters related to the risk-free component of the risky interest rates.

The state variable $y(t)$ addresses the stochastic behaviour of the credit spread and it is such that

$$d[Ly(t)] = Ldy(t) = -La_y y(t)dt + L\sigma_y dW_y(t), \quad y(0) = 0, \quad (13)$$

where a_y and σ_y are the model parameters related to the credit-risk sensitive component of the risky rates.

Under the two-factor model described above, the Brownian motions $dW_x(t)$ and $dW_y(t)$ are correlated such that

$$dW_x(t)dW_y(t) = \rho dt, \quad (14)$$

where the parameter ρ , with $-1 \leq \rho \leq 1$, represents the correlation coefficient between the risk-free interest rates and the credit spreads.

In the spirit of the model we propose, instead of calculating $\alpha(t)$ and $\beta(t)$ in a separate way, we calculate directly $\bar{\alpha}(t)$. Consequently, the function $\bar{\alpha}(t)$ is as follows,

$$\begin{aligned} \bar{\alpha}(t) = \bar{f}^M(0, t) + \frac{\sigma_x^2}{2a_x^2}(1 - e^{-a_x t})^2 + L^2 \frac{\sigma_y^2}{2a_y^2}(1 - e^{-a_y t})^2 + \\ L\rho \frac{\sigma_x \sigma_y}{a_x a_y}(1 - e^{-a_x t})(1 - e^{-a_y t}), \end{aligned} \quad (15)$$

where $\bar{f}^M(0, t)$ is the instantaneous forward risky interest rate derived by the market and prevailing at time 0 for the maturity t .

Pricing and stochastic dynamic for defaultable bonds under the two-factor Gaussian model

Defaultable zero-coupon bonds

The price of a defaultable zero-coupon bond at time t with maturity at time T can be expressed analytically as follows,

$$\bar{P}(t, T) = \frac{\bar{P}^M(0, T)}{\bar{P}^M(0, t)} \frac{P_{xy}(0, t)}{P_{xy}(0, T)} P_{xy}(t, T), \quad (16)$$

where $\bar{P}^M(0, T)$ is the observed market price and $P_{xy}(t, T)$ is such that

$$P_{xy}(t, T) = P_x(t, T)P_y(t, T)C_{x,y}(t, T), \quad (17)$$

with,

$$P_x(t, T) = G_x(t, T)e^{-H_x(t, T)x(t)}, \quad (18)$$

$$P_y(t, T) = G_y(t, T)e^{-H_y(t, T)Ly(t)}. \quad (19)$$

The quantity $C_{xy}(t, T)$ represents the correlation-related component of the price and it is formulated as follows,

$$C_{x,y}(t, T) = \exp \left\{ L \frac{\rho \sigma_x \sigma_y}{a_x a_y} \left[(T - t) - H_x(t, T) - H_y(t, T) + \frac{1}{a_x + a_y} \left(1 - e^{-(a_x + a_y)(T-t)} \right) \right] \right\}. \quad (20)$$

Furthermore, we define the following quantities,

$$G_x(t, T) = \exp \left\{ -\frac{\sigma_x^2}{2a_x^2} [H_x(t, T) - (T - t)] - \frac{\sigma_x^2}{4a_x} H_x(t, T)^2 \right\}, \quad (21)$$

$$G_y(t, T) = \exp \left\{ -L^2 \frac{\sigma_y^2}{2a_y^2} [H_y(t, T) - (T - t)] - L^2 \frac{\sigma_y^2}{4a_y} H_y(t, T)^2 \right\}, \quad (22)$$

$$H_x(t, T) = \frac{1}{a_x} \left[1 - e^{-a_x(T-t)} \right], \quad (23)$$

$$H_y(t, T) = \frac{1}{a_y} \left[1 - e^{-a_y(T-t)} \right]. \quad (24)$$

Following Schonbucker (2004), under the risk-neutral measure $\mathbb{Q}$, the price of a defaultable zero-coupon bond with maturity T satisfies the following stochastic differential equation,

$$\frac{d\bar{P}(t, T)}{\bar{P}(t, T)} = \bar{r}(t)dt - \sigma_x D_{P_x}(t, T)dW_x(t) - L\sigma_y D_{P_y}(t, T)dW_y(t). \quad (25)$$

The quantities $D_{P_x}(t, T)$ and $D_{P_y}(t, T)$ are stochastic durations of the zero-coupon bond with respect to the factors x and y respectively,

$$D_{P_x}(t, T) = -\frac{1}{\bar{P}(t, T)} \frac{\partial \bar{P}(t, T)}{\partial x(t)} = H_x(t, T), \quad (26)$$

$$D_{P_y}(t, T) = -\frac{1}{\bar{P}(t, T)} \frac{\partial \bar{P}(t, T)}{\partial y(t)} = H_y(t, T). \quad (27)$$

Defaultable coupon-bearing bonds

Let's consider a coupon-bearing bond with cash flows payments K_i at time T_i such that:

$$K_i = K\tau(T_{i-1}, T_i) \text{ for } i = 1, 2, \dots, n-1,$$

$$K_i = K\tau(T_{i-1}, T_i) + 1 \text{ for } i = n,$$

where K is the coupon rate and $\tau(T_{i-1}, T_i)$ denotes the time measure between T_{i-1} and T_i computed as a fraction of the year.

Denoting by $\bar{B}(t, T_n)$ the price at time t of a defaultable coupon-bearing bond with maturity T_n , we have that

$$\bar{B}(t, T_n) = \sum_{i=1}^n K_i \bar{P}(t, T_i). \quad (28)$$

In the spirit of Schonbucker (2004) and following Russo and Fabozzi (2016 and 2017), we assume that under the risk-neutral measure $\mathbb{Q}$ the market value of a defaultable coupon-bearing bond at time t evolves according to the following stochastic dynamic,

$$\frac{d\bar{B}(t, T_n)}{\bar{B}(t, T_n)} = \bar{r}(t)dt - \sigma_x D_{B_x}(t, T_n) dW_x(t) - L\sigma_y D_{B_y}(t, T_n) dW_y(t). \quad (29)$$

The quantities $D_{B_x}(t, T_n)$ and $D_{B_y}(t, T_n)$ are the stochastic durations of the defaultable coupon-bearing bond with respect to the factors x and y respectively,

$$D_{B_x}(t, T_n) = -\frac{1}{\bar{B}(t, T_n)} \frac{\partial \bar{B}(t, T_n)}{\partial x(t)} = \frac{\sum_{i=1}^n K_i \bar{P}(t, T_i) H_x(t, T_i)}{\sum_{i=1}^n C_i \bar{P}(t, T_i)}, \quad (30)$$

$$D_{B_y}(t, T_n) = -\frac{1}{\bar{B}(t, T_n)} \frac{\partial \bar{B}(t, T_n)}{\partial y(t)} = \frac{\sum_{i=1}^n K_i \bar{P}(t, T_i) H_y(t, T_i)}{\sum_{i=1}^n C_i \bar{P}(t, T_i)}. \quad (31)$$

Pricing options on defaultable bonds

A European call/put option on a defaultable bond gives the owner of this option the right to buy/sell the bond at the exercise date of the option for a given strike price. As stated by Schonbucker (2004), when analyzing the value of options having as their underlying defaultable securities, default risk should be not ignored as the option's value is composed of two parts: price protection component and default protection component. The price protection component is the value of the option if there is no default of the underlying bond. The default protection component is the value of the option at default of the underlying bond. The first component of the option's value is driven by normal price changes of the bond over the the option's life. The second component is either a default insurance (for

a put option on the bond) or zero (for a call option on the bond). This type of option protects the buyer against all risks: interest rate risk, credit spread risk or default risk that could move the defaultable bond price below the strike.

However, when the default protection component is considered, if bankruptcy occurs at any time before expiration, then the holder can immediately exercise the option. Under this assumption, the holder keeps the right to exercise the option in any case, since if default occurs contractual provisions entail the sudden expiration of the option itself. Consequently, if this assumption holds, then a closed-form solution could not be derived. Indeed, if options are protected, they cannot be defined as completely European, because if default occurs the holder does not have to wait until natural expiration to exercise his right.

It may seem that focusing just on the price protection component is of little practical interest. Nevertheless, it is important to establish a robust framework for evaluating the price protection component (i.e., the European option) as the starting point to evaluate the default protection component.⁴

In this paper, we assume unprotected options providing a framework for the evaluation of the price protection component where the option is assumed to be knocked out at default.

Let's denote by $ZC\bar{P}C(t, T_0, T_n, X)$ the price at time t of an unprotected European call option with strike X and exercise date $T_0 > t$, written on a defaultable zero-coupon bond with maturity $T_n > T_0$. Assuming that the bond has not defaulted by the valuation date t , the price of such option is,

$$ZC\bar{P}C(t, T_0, T_n, X) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_0} [r(u) + L\lambda(u)] du} \left(\bar{P}(T_0, T_n) - X \right)^+ \middle| \bar{\mathcal{R}}_t \right], \quad (32)$$

while the price of the corresponding put is,

$$ZC\bar{P}P(t, T_0, T_n, X) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_0} [r(u) + L\lambda(u)] du} \left(X - \bar{P}(T_0, T_n) \right)^+ \middle| \bar{\mathcal{R}}_t \right]. \quad (33)$$

When the underlying of the option is a defaultable coupon-bearing bond with maturity T_n , the price at time t of a knockout European call option with strike X and exercise date T_0 is,

$$CB\bar{B}C(t, T_0, T_n, X) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_0} [r(u) + L\lambda(u)] du} \left(\bar{B}(T_0, T_n) - X \right)^+ \middle| \bar{\mathcal{R}}_t \right], \quad (34)$$

while the pricing of the corresponding put is,

$$CB\bar{B}P(t, T_0, T_n, X) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_0} [r(u) + L\lambda(u)] du} \left(X - \bar{B}(T_0, T_n) \right)^+ \middle| \bar{\mathcal{R}}_t \right]. \quad (35)$$

When Monte Carlo simulation techniques are used to evaluate the expectations above, the risk-neutral measure $\mathbb{Q}$ can be used to evaluate the discounted payoff

⁴Barone et al. (1998).

and the price of the underlying bond can be simulated discretizing the stochastic processes described in the previous sections. However, in order to derive a closed-formula for such payoff, the correlation between the discounting and the payoff make the expectation difficult to evaluate. This difficulty can be overcome by changing the probability measure: as in the case of options written on default-free instruments, for the pricing of options on defaultable bonds, we adopt the $\mathbb{T}$ -forward survival risk-adjusted measure or survival measure ⁵ where $\mathbb{E}^{\mathbb{T}}$ denotes the expectation under this measure.

Options on defaultable zero-coupon bonds

Given the spot price of a defaultable zero-coupon bond, we define its forward price as,

$$\bar{P}(t, T_0, T_n) = \frac{\bar{P}(t, T_n)}{\bar{P}(t, T_0)}, \quad (36)$$

with $t < T_0 < T_n$. We assume the defaultable zero-coupon bond forward price is a martingale under the $\mathbb{T}$ -forward survival risk-adjusted measure.

Following Russo and Fabozzi (2017), by application of the Ito's lemma, we obtain that

$$\frac{d\bar{P}(t, T_0, T_n)}{\bar{P}(t, T_0, T_n)} = -\sigma_x D_{P_x}(t, T_0, T_n) dW_x^T(t) - L\sigma_y D_{P_y}(t, T_0, T_n) dW_y^T(t), \quad (37)$$

where $dW_x^T(t)$ and $dW_y^T(t)$ are Brownian motions under the $\mathbb{T}$ -forward survival risk-adjusted measure. The quantities $D_{P_x}(t, T_0, T_n)$ and $D_{P_y}(t, T_0, T_n)$ are defined as forward stochastic durations of the zero-coupon bond with respect to the factors x and y respectively. After simple calculations we obtain,

$$\begin{aligned} D_{P_x}(t, T_0, T_n) &= D_{P_x}(t, T_n) - D_{P_x}(t, T_0) = H_x(t, T_n) - H_x(t, T_0), \\ D_{P_y}(t, T_0, T_n) &= D_{P_y}(t, T_n) - D_{P_y}(t, T_0) = H_y(t, T_n) - H_y(t, T_0). \end{aligned}$$

Given the process above, the variance of the forward price of the defaultable zero-coupon bond is,

$$\begin{aligned} \bar{\sigma}_P(t, T_0, T_n)^2 &= \sigma_x^2 D_{P_x}(t, T_0, T_n)^2 + L^2 \sigma_y^2 D_{P_y}(t, T_0, T_n)^2 + \\ &\quad + 2L\rho\sigma_x\sigma_y D_{P_x}(t, T_0, T_n) D_{P_y}(t, T_0, T_n). \end{aligned}$$

In order to provide the pricing function, we need to derive the variance of the defaultable zero-coupon bond price under the $\mathbb{T}$ -forward survival risk-adjusted measure,

$$\bar{\Sigma}_P(t, T_0, T_n)^2 = \int_t^{T_0} \bar{\sigma}_P(u, T_0, T_n)^2 du. \quad (38)$$

⁵See Schombucher (2004).

Consequently, we find that the volatility of the defaultable zero-coupon bond is

$$\bar{\Sigma}_P(t, T_0, T_n) = \sqrt{\int_t^{T_0} \bar{\sigma}_P(u, T_0, T_n)^2 du}. \quad (39)$$

Solving the integral, it follows that,

$$\begin{aligned} \bar{\Sigma}(t, T_0, T_n)^2 &= \frac{\sigma_x^2}{2a_x^3} \left[1 - e^{-a_x(T_n - T_0)} \right]^2 \left[1 - e^{-2a_x(T_0 - t)} \right] + \\ &L^2 \frac{\sigma_y^2}{2a_y^3} \left[1 - e^{-a_y(T_n - T_0)} \right]^2 \left[1 - e^{-2a_y(T_0 - t)} \right] + \\ &2L\rho \frac{\sigma_x \sigma_y}{a_x a_y (a_x + a_y)} \left[1 - e^{-a_x(T_n - T_0)} \right] \left[1 - e^{-a_y(T_n - T_0)} \right] \left[1 - e^{-(a_x + a_y)(T_0 - t)} \right]. \end{aligned} \quad (40)$$

Assuming that the defaultable zero-coupon is approximately log-normal and assuming that risky rates are stochastic and evolve according to the two-factor Gaussian model, the arbitrage-free price for an option written on a defaultable zero-coupon bond is,

$$ZC\bar{P}C(t, T_0, T_n, X) = \bar{P}(t, T_0) \mathbb{E}^{\mathbb{T}} \left[\left(\bar{P}(T_0, T_n) - X \right)^+ \middle| \bar{\mathcal{R}}_t \right]. \quad (41)$$

Equation (41) is valid under the survival measure as we implicitly assume the zero-coupon defaultable bond maturing at time T_0 to be the numeraire. Consequently, we have that

$$ZC\bar{P}C(t, T_0, T_n, X) = \bar{P}(t, T_n) \Phi(d_1) - X \bar{P}(t, T_0) \Phi(d_2), \quad (42)$$

where Φ denotes the cumulative distribution function of the standard Gaussian distribution with,

$$d_1 = \frac{\log \left[\frac{\bar{P}(t, T_n)}{\bar{P}(t, T_0)} \frac{1}{X} \right] + \frac{1}{2} \bar{\Sigma}_P(t, T_0, T_n)^2}{\bar{\Sigma}_P(t, T_0, T_n)}, \quad (43)$$

and

$$d_2 = \frac{\log \left[\frac{\bar{P}(t, T_n)}{\bar{P}(t, T_0)} \frac{1}{X} \right] - \frac{1}{2} \bar{\Sigma}_P(t, T_0, T_n)^2}{\bar{\Sigma}_P(t, T_0, T_n)}. \quad (44)$$

The corresponding value of the zero-coupon bond put option, denoted by $ZC\bar{P}P(t, T_0, T_n, X)$, can be obtained by simply applying the put-call parity.

Options on defaultable coupon-bearing bonds

As for the case of the option on zero-coupon bond, given the spot price of a defaultable coupon-bearing bond, we define its forward price as,

$$\bar{B}(t, T_0, T_n) = \frac{\bar{B}(t, T_n)}{\bar{P}(t, T_0)}, \quad (45)$$

with $t < T_0 < T_n$. We assume the defaultable coupon-bearing bond forward price is a martingale under the $\mathbb{T}$ -forward survival risk-adjusted measure.

Also in this case, following Russo and Fabozzi (2017) and by application of the Ito's lemma, we obtain that

$$\frac{d\bar{B}(t, T_0, T_n)}{\bar{B}(t, T_0, T_n)} = -\sigma_x D_{B_x}(t, T_0, T_n) dW_x^T(t) - L\sigma_y D_{B_y}(t, T_0, T_n) dW_y^T(t), \quad (46)$$

where $dW_x^T(t)$ and $dW_y^T(t)$ are Brownian motions under the $\mathbb{T}$ -forward survival risk-adjusted measure. The quantities $D_{B_x}(t, T_0, T_n)$ and $D_{B_y}(t, T_0, T_n)$ are defined as forward stochastic durations of the coupon-bearing bond with respect to the factors x and y respectively. After simple calculations we obtain,

$$D_{B_x}(t, T_0, T_n) = D_{B_x}(t, T_n) - D_{B_x}(t, T_0) = \frac{\sum_{i=1}^n K_i \bar{P}(t, T_i) [H_x(t, T_i) - H_x(t, T_0)]}{\sum_{i=1}^n K_i \bar{P}(t, T_i)},$$

$$D_{B_y}(t, T_0, T_n) = D_{B_y}(t, T_n) - D_{B_y}(t, T_0) = \frac{\sum_{i=1}^n K_i \bar{P}(t, T_i) [H_y(t, T_i) - H_y(t, T_0)]}{\sum_{i=1}^n K_i \bar{P}(t, T_i)}.$$

Given the process above, the variance of the forward price of the defaultable coupon-bearing bond is,

$$\begin{aligned} \bar{\sigma}_B(t, T_0, T_n)^2 &= \sigma_x^2 D_{B_x}(t, T_0, T_n)^2 + L^2 \sigma_y^2 D_{B_y}(t, T_0, T_n)^2 + \\ &\quad + 2L\rho\sigma_x\sigma_y D_{B_x}(t, T_0, T_n) D_{B_y}(t, T_0, T_n). \end{aligned}$$

In order to provide the pricing function, we need to derive the variance of the defaultable coupon-bearing bond price under the $\mathbb{T}$ -forward survival risk-adjusted measure,

$$\bar{\Sigma}_B(t, T_0, T_n)^2 = \int_t^{T_0} \bar{\sigma}_B(u, T_0, T_n)^2 du. \quad (47)$$

Consequently, we find that the volatility of the defaultable coupon-bearing bond is

$$\bar{\Sigma}_B(t, T_0, T_n) = \sqrt{\int_t^{T_0} \bar{\sigma}_B(u, T_0, T_n)^2 du}. \quad (48)$$

A numerical integration method is required here to calculate the volatility of the defaultable coupon-bearing bond under the proposed model.

Assuming that the defaultable coupon-bearing bond is approximately log-normal and assuming that risky rates are stochastic and evolve according to a two-factor Gaussian model, the arbitrage-free price for an option written on a defaultable coupon-bearing bond is,

$$CB\bar{B}C(t, T_0, T_n, X) = \bar{B}(t, T_0) \mathbb{E}^{\mathbb{T}} \left[\left(\bar{B}(T_0, T_n) - X \right)^+ \middle| \bar{\mathcal{H}}_t \right]. \quad (49)$$

Consequently, we have that

$$CB\bar{B}C(t, T_0, T_n, X) = \bar{B}(t, T_n) \Phi(d_1) - X \bar{P}(t, T_0) \Phi(d_2), \quad (50)$$

where,

$$d_1 = \frac{\log \left[\frac{\bar{B}(t, T_n)}{\bar{P}(t, T_0)} \frac{1}{X} \right] + \frac{1}{2} \bar{\Sigma}_B(t, T_0, T_n)^2}{\bar{\Sigma}_B(t, T_0, T_n)}, \quad (51)$$

and

$$d_2 = \frac{\log \left[\frac{\bar{B}(t, T_n)}{\bar{P}(t, T_0)} \frac{1}{X} \right] - \frac{1}{2} \bar{\Sigma}_B(t, T_0, T_n)^2}{\bar{\Sigma}_B(t, T_0, T_n)}. \quad (52)$$

The corresponding value of the coupon-bearing bond put option, denoted by $CB\bar{B}P(t, T_0, T_n, X)$, can be obtained by simply applying the put-call parity.

Model's calibration under a market-consistent approach

In order to perform a market-consistent calibration of the proposed model for a specific date, the following information must be obtained:⁶

- the term structure of risk-free interest rates;
- the at-the-money (ATM) quotation of swaptions to be used for the calibration of parameters related to the risk-free rates model component;
- the term structure of defaultable interest rates in relation to a particular issuer, and;
- the market consensus recovery rate (RR) and the related loss given default (L), with $RR = 1 - L$.

⁶For all quantities observed in the market we use the suffix M .

The calibration is performed using the following two-step procedure:

- Step 1: calibration of the model, in relation to the risk-free rate process, performed using swaptions;
- Step 2: calibration of the model, in relation to the credit spread process, performed using the term structure of defaultable interest rates in relation to a particular issuer.

In the first step, the calibration of the parameters a_x and σ_x is performed. For this step, we take into consideration the one-factor Hull-White model⁷ that explains the dynamic of the short risk-free rate in a standalone manner. In particular, neglecting the credit spread component, we refer to Russo and Fabozzi (2016) where an alternative representation of the Hull-White model is described according to the following dynamic,

$$r(t) = \alpha(t) + x(t), \quad (53)$$

where $x(t)$ is the state variable while $\alpha(t)$ is a deterministic function of time. The variable $x(t)$ is such that

$$dx(t) = -a_x x(t)dt + \sigma_x dW_x(t), \quad x(0) = 0, \quad (54)$$

where a_x and σ_x are model parameters while $dW_x(t)$ is a Brownian motion.

In order to perform the calibration, we use ATM swaptions quoted by the market. Following Russo and Fabozzi (2016), the calibration to swaptions is performed by choosing the values of the parameters so as to minimize the square root of the sum of the squares of the relative differences between market and model swaption prices,

$$\arg \min_{a_x, \sigma_x} \sqrt{\sum_{i=1}^{N_1} \left(\frac{Swpt_i - Swpt_i^M}{Swpt_i^M} \right)^2}, \quad (55)$$

where $Swpt_i^M$ is the value of the swaption quoted by the market and $Swpt_i$ represents the swaption's theoretical price under the Hull-White model. The number of calibrated instruments is N_1 .

The second step involves the calibration of the model's component related to the credit spread. In this case, a_y and σ_y , and the correlation coefficient ρ have to be calibrated. It is worth noting that the second calibration step is performed with reference to a particular issuer, taking into account its specific credit spread and loss given default L .

Unfortunately, a liquid market of options having as its underlying defaultable

⁷Hull and White (1990).

bonds is not available. If such instruments would be quoted in the market, it would be possible to calibrate the parameters of the credit's component of our model in a similar manner in which swaptions are used to calibrate the risk-free part.

In order to overcome this issue, we use market prices of defaultable bonds to perform the second calibration procedure. Given the parameters estimated under the first step, we minimize the square root of the sum of the squares of the relative differences between market and model prices related to a set (N_2) of defaultable zero-coupon bonds,

$$\arg \min_{a_y, \sigma_y, \rho} \sqrt{\sum_{i=1}^{N_2} \left(\frac{\bar{P}^*(t, T_i) - \bar{P}(t, T_i)^M}{\bar{P}(t, T_i)^M} \right)^2}, \quad (56)$$

where $\bar{P}(t, T_i)^M$ is the value of the defaultable zero-coupon bond price quoted by the market while $\bar{P}^*(t, T_i)$ represents the theoretical price of a defaultable zero-coupon bond calculated numerically as follows,

$$\bar{P}^*(t, T_i) = \exp \left\{ - \frac{\sum_{j=1}^i T_j \bar{\alpha}(T_j)}{T_i} \right\}. \quad (57)$$

To evaluate the quality fitting, we calculate the Root Mean Square Error (RMSE), using market and model prices, computed as in Equation (56).

In order to obtain the parameters of the model's credit component, the optimization procedure described above has to be performed several times. In fact, such procedure is performed under different assumptions about the correlation between the two model factors $x(t)$ and $y(t)$. We repeat the optimization procedure changing the correlation assumption at each iteration. As the correlation parameter is in the range $[-1; 1]$, we discretize this range to define a finite set of correlation figures to be evaluated across the calibration procedure. The final set of parameters in terms of a_y , σ_y , and ρ is found in correspondence of that specific iteration where the RMSE results as the minimum across all the iterations considered under the optimization procedure.

Empirical analysis

Calibration results

In order to perform the calibration on real market data, we referred to U.S. financial market with a point-in-time set of data obtained from Bloomberg for September 21, 2017. The calibration of the risk-free component of the model was performed using USD swaption prices. We used at-the-money (ATM) co-terminal swaptions because it is a common practice to calibrate interest rate models, in part for hedging reasons. Then, we considered a set of risky yield curves across different sectors

Exhibit 3: Financial sector - calibration results

Rating	a_x	σ_x	a_y	σ_y	ρ	RMSE
AAA	4.81%	0.97%	23.27%	9.68%	-10%	0.76%
AA	4.81%	0.97%	25.00%	10.11%	0%	0.75%
A	4.81%	0.97%	23.73%	10.38%	-10%	1.68%
BBB	4.81%	0.97%	17.96%	10.44%	-50%	2.63%
BB	4.81%	0.97%	26.52%	12.98%	40%	3.99%
B	4.81%	0.97%	30.94%	14.17%	80%	4.99%

and credit ratings. In particular, we considered financial and industrial sector yield curves for the following credit ratings: AAA, AA, A, BBB, BB, and B. In a coherent manner with the assumption underlying the quotation of standardized credit default swap (CDS) contracts, we assume a RR equal to 40% for all the cases considered.

In Exhibits 1 and 2, we show the yield curves related respectively to the financial and industrial sector across the different credit ratings considered. In both exhibits, the USD risk-free yield curve is also reported.

Exhibit 1: Financial sector - Yield curves as of September 21, 2017

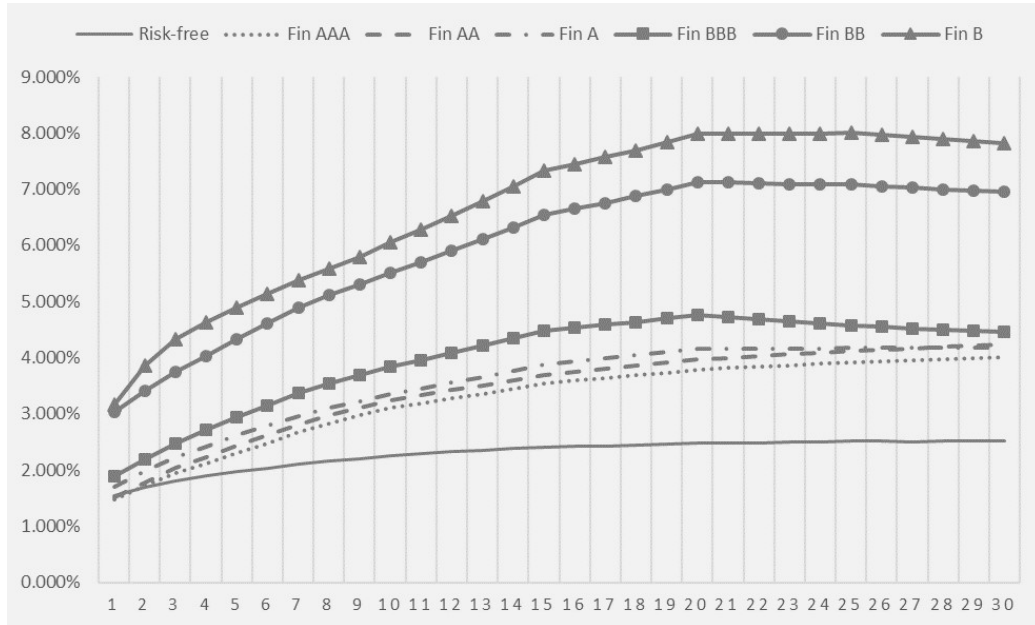


Exhibit 2: Industrial sector - Yield curves as of September 21, 2017

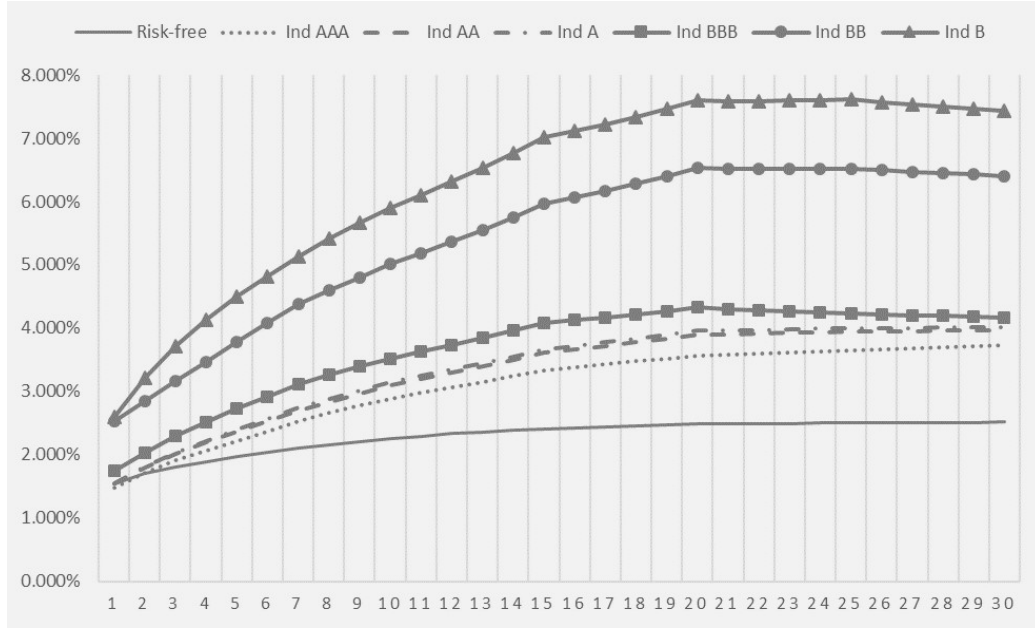
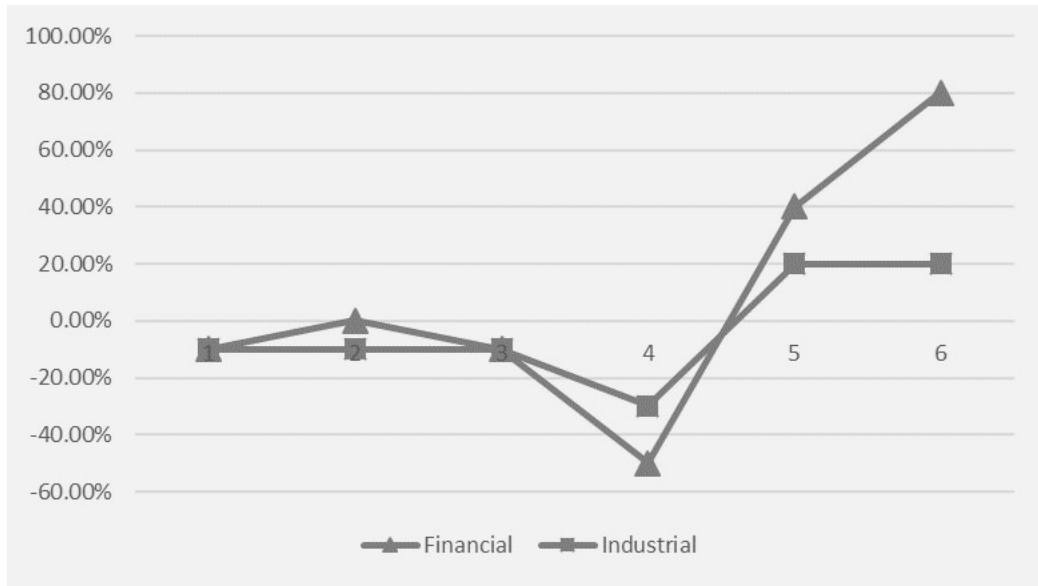


Exhibit 4: Industrial sector - calibration results

Rating	a_x	σ_x	a_y	σ_y	ρ	RMSE
AAA	4.81%	0.97%	24.53%	9.77%	-10%	1.13%
AA	4.81%	0.97%	22.76%	9.59%	-10%	1.48%
A	4.81%	0.97%	22.67%	9.64%	-10%	1.51%
BBB	4.81%	0.97%	20.89%	10.40%	-30%	2.11%
BB	4.81%	0.97%	22.56%	11.35%	20%	3.22%
B	4.81%	0.97%	22.15%	12.27%	20%	3.54%

Note from Exhibits 3 and 4 that while the parameters of the risk-free component of the model are unchanged (a_x and σ_x), the other parameters related to the credit component (a_y , σ_y and ρ) vary across the different credit ratings. In particular, it can be seen that the lower the credit rating the higher the parameter σ_y . In addition, the lower the credit rating the higher the correlation between the risk-free rates and the spread as can be seen in Exhibit 5.

Exhibit 5: Correlations for the different credit rating (From AAA=1 to B=6).



Option pricing results

In this section, we show the results of the proposed model in terms of options pricing using the calibration results reported in the previous section. We consider options written on defaultable zero-coupon bonds (ZCB) and options written on defaultable coupon-bearing bonds (CBB) across different credit ratings (AAA, AA, A, BBB, BB, B) and across different assumptions in terms of loss given default L ($L = 20\%$, $L = 40\%$, $L = 60\%$, $L = 80\%$). The face value of the underlying bonds is assumed to be equal to 1 unit of currency. For both type of underlyings across all the credit ratings under analysis, we consider similar swaptions with a maturity equal to five years and a bond maturity equal to 15 years.⁸ Moreover, for the options written on defaultable zero-coupon bonds, we assume the strike is equal to 50% of the face value, while for the options written on defaultable coupon-bearing

⁸For sake of clarity, to be highlighted that the bond's maturity starts from the valuation date t (i.e., September 21, 2017).

bonds the strike is assumed to be equal to 100% of the face value.

In the four panels in Exhibit 6, the prices of options written on defaultable zero-coupon bonds are reported, respectively for both financial and industrial sectors and for both call and put options. Similar results for options written on defaultable coupon-bearing bonds are reported across the four panels in Exhibit 7. Such results have been obtained under the proposed closed formulas.

In order to verify the correctness of the results, we have tested the consistency of the prices in terms of the put-call parity under defaultable securities. Following Fabozzi (2001), the put-call parity in case of options having defaultable ZCBs as underlying is the following,

$$ZC\bar{P}P(t, T_0, T_n, X) = ZC\bar{P}C(t, T_0, T_n, X) + X\bar{P}(t, T_0) - \bar{P}(t, T_n). \quad (58)$$

Analogously, the put-call parity in case of options having defaultable CBBs as their underlying becomes

$$\begin{aligned} CB\bar{B}P(t, T_0, T_n, X) &= CB\bar{B}C(t, T_0, T_n, X) + X\bar{P}(t, T_0) - \bar{B}(t, T_n) + \\ &+ \bar{B}(t, T_0) - \bar{P}(t, T_0). \end{aligned} \quad (59)$$

Applying the two conditions reported above, all the results are consistent. Moreover, for testing purposes, we have calculated the same prices using a Monte Carlo simulation approach and we have obtained results very close to those obtained with the proposed model. We recalculated the option prices using 25,000 simulations. In all the cases considered, we find that the mean absolute relative difference is less than 2%.

Looking at the results obtained, it is important to highlight that prices are consistent also in terms of the credit risk embedded in the underlying. On the one hand, it has to be noted that the higher the loss given default L , the higher the option's price for both call and put options. On the other hand, the put values increase and the call values decrease as the creditworthiness of the underlying decreases.⁹

It is important to highlight that when the loss given default increases, on the same rating, the volatility of the underlying increases; at the same time, the numeraire, affecting the discounting of the option, remains the same as the numeraire is equal to the market price of the defaultable bond. Instead, when the rating decreases, under the same loss given default, the volatility of the underlying increases as the creditworthiness of the underlying decreases; however, at the same time, the

⁹These relations are generally valid with some expected exceptions related to particular combinations of rating and loss given default in case of CBBs. Such special cases are essentially due to the presence of the coupon and to the discounting effect under the chosen measure.

numeraire, affecting the discounting of the option, changes according to the change in credit rating.

Conclusion

In this paper, we provide a closed-form solution for defaultable bond options under a two-factor Gaussian model for risky rates. The key feature of the proposed stochastic model is the introduction of two stochastic dynamics to address the behavior of both risk-free interest rates and credit spreads where the two sources of risk are correlated. Moreover, the model is able to match exactly the term structure of defaultable interest rates. In order to model credit-sensitive bonds, we assume an intensity-based reduced-form framework where the approach based on the recovery of market value is considered and where the recovery rate is one of the model's input. Under this framework, we derive closed formulas for the price of European options having as their underlying both defaultable zero-coupon bonds and coupon-bearing bonds which we assume are approximately log-normally distributed. Moreover, the options are assumed to be knocked out upon a default event of the bond with zero value to the option holder. An empirical analysis illustrates how to perform the calibration process of the proposed model using financial market data to price calls and puts options across different credit rating and different levels of the loss given default.

Disclaimer: Vincenzo Russo and not his employer is solely responsible for any errors.

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Exhibit 6: Call and put options on defaultable zero-coupon bonds

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	0.00%	0.2327	0.0968	-10%	0.1443	0.1503	0.1616	0.1764
AA	0.00%	0.2500	0.1011	0%	0.1351	0.1418	0.1533	0.1677
A	0.00%	0.2373	0.1038	-10%	0.1235	0.1314	0.1447	0.1610
BBB	0.00%	0.1796	0.1044	-50%	0.0837	0.0974	0.1189	0.1430
BB	0.00%	0.2652	0.1298	40%	0.0230	0.0383	0.0549	0.0721
B	0.00%	0.3094	0.1417	80%	0.0136	0.0260	0.0396	0.0537

Panel a. Financial sector: call options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	0.00%	0.2327	0.0968	-10%	0.0023	0.0084	0.0197	0.0344
AA	0.00%	0.2500	0.1011	0%	0.0033	0.0100	0.0215	0.0359
A	0.00%	0.2373	0.1038	-10%	0.0037	0.0116	0.0249	0.0412
BBB	0.00%	0.1796	0.1044	-50%	0.0049	0.0187	0.0402	0.0642
BB	0.00%	0.2652	0.1298	40%	0.0514	0.0666	0.0833	0.1005
B	0.00%	0.3094	0.1417	80%	0.0722	0.0846	0.0983	0.1123

Panel b. Financial sector: put options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	0.00%	0.2453	0.0977	-10%	0.1600	0.1645	0.1740	0.1870
AA	0.00%	0.2276	0.0959	-10%	0.1399	0.1463	0.1582	0.1734
A	0.00%	0.2267	0.0964	-10%	0.1364	0.1432	0.1554	0.1710
BBB	0.00%	0.2089	0.1040	-30%	0.1100	0.1197	0.1361	0.1557
BB	0.00%	0.2256	0.1135	20%	0.0333	0.0501	0.0688	0.0880
B	0.00%	0.2215	0.1227	20%	0.0145	0.0299	0.0478	0.0665

Panel c. Industrial sector: call options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	0.00%	0.2453	0.0977	-10%	0.0015	0.0061	0.0155	0.0286
AA	0.00%	0.2276	0.0959	-10%	0.0026	0.0091	0.0209	0.0361
A	0.00%	0.2267	0.0964	-10%	0.0028	0.0097	0.0219	0.0374
BBB	0.00%	0.2089	0.1040	-30%	0.0037	0.0134	0.0298	0.0494
BB	0.00%	0.2256	0.1135	20%	0.0382	0.0550	0.0737	0.0929
B	0.00%	0.2215	0.1227	20%	0.0647	0.0801	0.0980	0.1167

Panel d. Industrial sector: put options

Exhibit 7: Call and put options on defaultable coupon-bearing bonds

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	3.54%	0.2327	0.0968	-10%	0.0333	0.0549	0.0822	0.1115
AA	3.70%	0.2500	0.1011	0%	0.0346	0.0560	0.0822	0.1103
A	3.88%	0.2373	0.1038	-10%	0.0332	0.0561	0.0845	0.1147
BBB	4.48%	0.1796	0.1044	-50%	0.0215	0.0508	0.0884	0.1277
BB	6.55%	0.2652	0.1298	40%	0.0300	0.0555	0.0834	0.1119
B	7.34%	0.3094	0.1417	80%	0.0319	0.0562	0.0815	0.1069

Panel a. Financial sector: call options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	3.54%	0.2327	0.0968	-10%	0.0815	0.1031	0.1304	0.1597
AA	3.70%	0.2500	0.1011	0%	0.0826	0.1039	0.1302	0.1582
A	3.88%	0.2373	0.1038	-10%	0.0800	0.1029	0.1313	0.1615
BBB	4.48%	0.1796	0.1044	-50%	0.0757	0.1050	0.1426	0.1820
BB	6.55%	0.2652	0.1298	40%	0.0946	0.1200	0.1480	0.1765
B	7.34%	0.3094	0.1417	80%	0.0947	0.1190	0.1444	0.1698

Panel b. Financial sector: put options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	3.34%	0.2453	0.0977	-10%	0.0350	0.0554	0.0816	0.1098
AA	3.61%	0.2276	0.0959	-10%	0.0337	0.0557	0.0834	0.1131
A	3.67%	0.2267	0.0964	-10%	0.0335	0.0558	0.0837	0.1136
BBB	4.08%	0.2089	0.1040	-30%	0.0284	0.0539	0.0864	0.1209
BB	5.96%	0.2256	0.1135	20%	0.0287	0.0548	0.0845	0.1151
B	7.02%	0.2215	0.1227	20%	0.0261	0.0533	0.0840	0.1153

Panel c. Industrial sector: call options

Rating	Coupon	a_y	σ_y	ρ	$L = 20\%$	$L = 40\%$	$L = 60\%$	$L = 80\%$
AAA	3.34%	0.2453	0.0977	-10%	0.0786	0.0991	0.1252	0.1535
AA	3.61%	0.2276	0.0959	-10%	0.0807	0.1027	0.1304	0.1601
A	3.67%	0.2267	0.0964	-10%	0.0808	0.1031	0.1310	0.1609
BBB	4.08%	0.2089	0.1040	-30%	0.0776	0.1030	0.1355	0.1700
BB	5.96%	0.2256	0.1135	20%	0.0957	0.1218	0.1515	0.1821
B	7.02%	0.2215	0.1227	20%	0.0969	0.1242	0.1548	0.1861

Panel d. Industrial sector: put options

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